

Lab 2 Assignment

ECE 525.442 FPGA Design using VHDL

Simple Seven-Segment Display Controller

For this exercise, you will re-implement the requirements for Lab 1 but will use processes to describe the conditionals in your design. The goals for this lab are to familiarize yourself with using processes to describe the combinational circuits and to perform a detailed analysis of the device resource utilization. Refer to Lab 1 description for the details on the requirements for this Lab assignment.

Comparison

Make sure you have completed the design for Lab 2 and verified the design works properly on the Nexy's development board. Compare the device utilization for Lab 1 vs. the device utilization for this lab assignment. For simplicity, tabulate the resources utilized for each design using a table similar to the one below.

Resources	Lab 1	Lab 2
Slice LUTs as Logic		
Slice LUTs as Memory		
Slice Registers as Flip Flop		
Slice Registers as Latch		
Number of bonded IOBs		
Number of Clocks		

Note that some of the resource listed in the table above (highlighted in red) may not be reported in the device utilization summary. If the synthesis tools report that these resources have indeed been utilized for your implementation of Lab 2, then you may have an implementation error. Either correct this error or justify why the resources differ between the two lab assignments.

An alternate method for comparing the implementations for Lab 1 and Lab 2 is by observing the RTL schematics generated by the synthesis tools after the initial synthesis steps. The RTL schematics should be very similar or even match exactly if the implementations for both designs are equivalent. Once again, if they differ, carefully check the VHDL code for your implementation of lab 2 and correct any potential implementation errors, or justify why the RTL schematics differ between the two lab assignments.

To Submit

Files to Submit (minimum), zipped together as Lab2_<last_name>.zip:

- VHD, XDC, BIT files for Lab 2



- Complete Synthesis Results
- RTL schematic for Lab 1 and RTL schematic for Lab 2 (take a snapshot or save PDF)
- Report comparing resource utilization between Lab 1 and Lab 2 with discussion on utilization and RTL schematics

Note: The Grading sheet provide on the next page provides a good overview of the key points of the design and what the instructor will look for when grading. Please note that this is a guide and isn't comprehensive. Please read over the lab assignment carefully to verify all the proper documents are submitted.



Grading Sheet

Lab 2/Module 2: Simple 7-segment display w/ Processes

Student:

Grade	Out of	Notes
	100	
	25	Seven segment encoder module correctly implemented
	5	Value of SW3-SW0 shows up on seven-segment display
	10	Values displayed on seven segment display should only be hexadecimal
	10	Green LEDs light up according to which slider switch is in HIGH position
	25	Correct Operation of Push Buttons Slider switches SW3-SW0 represent input digit 1. With no buttons pressed, the digit is displayed in any display that has the corresponding slider switch set HIGH from SW11 to SW4. 2. When the UP BTN is pressed (BTNU), only the upper four displays will be active and showing the SW3 to SW0 value 3. When the DOWN BTN is pressed (BTND), only the bottom four displays will be active and showing the SW3 to SW0 value 4. When the CENTER BTN is pressed (BTNC), all of the displays will be active and showing all zeros
	10	Well Documented and appropriate VHDL Code
	15	Synthesis report with device resource utilization summary --Device resource utilization comparison for Lab 1 vs. Lab 2 implementation --RTL schematics for Lab 1 compared to Lab 2 --Explanation of the comparisons between the two implementations

